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End Semester Examination 2024

Name of the Program: B.Tech (EE) | Semester: IV

Name of the Course: Mathematics III | Course Code: **TMA 401**

Time: 3 Hours | Maximum Marks: 100

Note:

(i) All Questions are compulsory

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty

(iv) Each question carries 10 marks

Q1 **(20 marks)**

a) Three machine A, B, C produce respectively 60%, 30% , 10% of the total number of item of factory. The percentages of defective output of this machine are respectively 2%, 3%, 4%. An item is selected at random from the product and is found defective . Find the probability that item was produced by machine C.

b) An Urn Contain 10white, 15red and 25 black marbles. Determine the probability of drawing
(a) Either a white or a red marble.
(b)Either a white or a red or a black marble.

c) The number of airplane landings at an airport in a minute (X) and their probabilities are given by

X_i	0	1	2	3	4	5	6
$p(x_i)$	0.02	0.15	0.22	0.26	0.17	0.14	0.04

Find the mean and standard deviation of X.

CO 1

Q2 **(20 marks)**

a) The diameter of an water pipe, say X , is assumed to be a continuous random variable with probability density function $f(x) = 6x(1-x), 0 \le x \le 1$. **(i)** Check validity of f(x) to be *p.d.f.* **(ii)** Determine a number b such that $P(X < b) = P(X > b)$.

b) If X is a Poisson distribution variate such that $P(X=2) = 9P(X=4) + 90P(X=6)$ Find λ (the mean of X).

c) The random variable X has a Poisson distribution. If $P(X=1) = 0.01487$, $P(X=2) = 0.04461$. Then find $P(X=3)$.

CO 2

Q3 **(20 marks)**

a) The joint density of X and Y is $f(x,y) = \frac{x+2y}{27}$ where x and y can be assumed only integer value 0, 1, and 2. Find the conditional distribution of Y for X=x.

b) If $f(x,y) = \begin{cases} e^{-(x+y)}, & x \ge 0, y \ge 0 \\ 0, & elsewhere \end{cases}$ is joint probability density function of random variable X and Y, Find

(i) P[X<1]

(ii) P[X+Y<1]

CO 3

c) If X and Y have join p.d.f

Y \ X	1	2	3
1	0.1	0.1	0.2
2	0.2	0.3	0.1

Find

(i) Marginal distribution of X and Y

(ii) Conditional distribution of X given Y=1

(iii) Evaluate $P[x + y < 4]$

Q4 **(20 marks)**

a) Toss 3 coins. Let X denotes the number of heads on the first two and Y denotes the Number of heads on the last two.

(i) Find the joint distribution of X and Y

(ii) Find $E[y / x = 1]$

(iii) Find the correlation coefficient $\rho(x, y)$

b) An Automobile tyres manufacturing claims that the average life of a particular grade of tyres is more than 20,000 kms when used under normal driving condition. A random sample of 16 tyres was tested, mean and standard deviation of 2200 and 5000 kms respectively were computed. Assuming the life of the tyres in kms to be normally distributed, decided whether the manufacturer's claim is true? At 5% level of significance, the critical value is 1.75 **CO 4**

c) The following figure relate to the number of limit producer per shift by two workers A and B a number of day:

A	19	22	24	27	24	18	20	19	25		
B	26	37	40	35	30	30	40	26	30	35	45

Can it be inferred that A is more stable worker compared to B ? Given that at 5% level of significance the critical value is 3.35

Q5 **(20 marks)**

a) To Study the relationship between **Aptitude test scores** and **productivity** resulting after three months training of workers in a factory, a survey was conducted by a researcher, who had taken a sample of 9 workers.

Aptitude Score	1	3	5	7	9	2	4	6	8
Productivity	9	10	11	14	15	8	12	13	16

Calculate coefficient of correlation between **Aptitude test scores** and **productivity** and give your comments.

b) 100 Students of a management institute obtained the following grades in statistics paper: **CO 5**

Grade	A	B	C	D	E	Total
Frequency	15	17	30	22	16	100

Examine the hypothesis that the distribution of grades is uniform. Given that at 5% level of significance the critical value is 9.49

c) From the following data determine line of regression equations and also calculate Correlation coefficient.

x	1	2	3	4	5	6	7	8
y	3	7	10	12	14	17	20	24

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End Semester Examination March 2024

Name of the Program: B.Tech
Name of the Course: Electrical Safety

Semester: 4th
Course Code: VAC-401

Time: 3 Hours

Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain 1. GLCB 2. MCCB 3. EMC 4. HRC	**CO 4**
(b)	Explain the effect of current leakage on a human body, depicting the zone-wise effect	
(c)	Explain the ground fault protection concept in hazardous areas	
Q2	(20 marks)	
(a)	Discuss the safety standards of electricity in detail, which one is associated with generator protection.	
(b)	What is insulation breakdown, give your point of view regarding electrical safety on the construction side.	**CO 2**
(c)	Explain the effect of surge on the electrical system. What is the difference between surge and Arc?	
Q3	(20 marks)	
(a)	Explain the role of isolators. Also, explain how they are different from circuit breakers.	
(b)	Discuss the ground fault protection schemes.	**CO 3**
(c)	What are CBCT?	
Q4	(20 marks)	
(a)	Explain 1. Electrostatics 2. Electromagnetism	**CO 1,2**
(b)	Discuss in brief the role of the Indian Electricity Act and explain any one in detail.	
(c)	Explain the concept of stored energy. Also, explain energy radiation.	
Q5	(20 marks)	
(a)	What is insulation breakdown, give your point of view regarding electrical safety at the domestic site.	**CO 4,1**
(b)	Explain the harmful effect of current on the human body. What is the implication of resistance of the human body?	

(c)	Explain the working principle of electrical equipment.	

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TEE–402

B. TECH. (EEE) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

INTRODUCTION TO ELECTRICAL ENERGY SOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate between fixed and operating costs of power plants. Which type of plant has highest capital cost ? Why ? (CO1)

   (b) Discuss the Flue gas cycle of a coal-based steam plant. (CO1)

   (c) Discuss why steam plants use pulverized coal. (CO1)

2. (a) Give the main features of Pressurized Water Reactor. Draw suitable diagram. (CO2)

   (b) Define Mass defect. Show that mass defect of 1 amu is equivalent to about 931 MeV of energy. (CO2)

   (c) How are nuclear reactors controlled ? (CO2)

3. (a) What is hydrograph ? What is the function of a surge tank ? (CO3)

   (b) The total water inflows for different months for a river are as under : (CO3)

   Jan. $1000*10^6$ m^3 **July** $3000*10^6$ m^3

P. T. O.

Feb.	$800*10^6$ m^3	**Aug.**	$3000*10^6$ m^3
March	$550*10^6$ m^3	**Sep.**	$2500*10^6$ m^3
April	$300*10^6$ m^3	**Oct.**	$1500*10^6$ m^3
May	$300*10^6$ m^3	**Nov.**	$1450*10^6$ m^3
June	$1500*10^6$ m^3	**Dec.**	$1600*10^6$ m^3

(i) Plot hydrograph and flow duration curve. Find out average monthly flow.

(ii) Find power available at mean flow if head available in 100 m and efficiency of generation is 80%.

(c) Differentiate between the operation of a Pelton turbine and a Francis turbine.

4. (a) Discuss the main parts of a diesel power plant. (CO4, CO5)

(b) Previously diesel and gas turbine plants were used only for peaking duty. Now they are also used as base load plants. Why ? (CO4, CO5)

(c) Discuss the working of Closed Cycle Gas Turbine plant with the help of suitable diagram. (CO4, CO5)

5. (a) Define load factor. Discuss its importance. (CO5, CO6)

(b) What objectives should a utility keep in m while deciding the tariff for consumers ? (CO5, CO6)

(c) The load on a power plant on a typical day is as under : (CO5, CO6)

Time	12-5 a.m.	5-9 a.m.	9-6 p.m.	6-10 p.m.	10-12 a.m.
Load (MW)	20	40	80	100	20

Plot the chronological load curve and load duration curve. Find the load factor of the plant and the energy supplied by plant in 24 hours.

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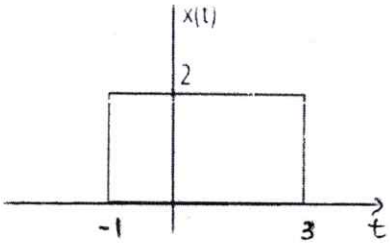
End Semester Examination 2024

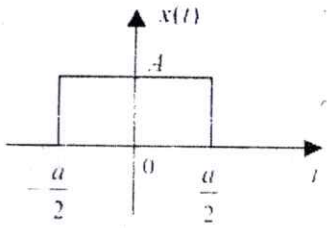
Name of the Course: B. Tech, EE — Semester: 4
Name of the Paper – Signals & Systems
Paper Code: TEE 403

Time: 3Hours — Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Discuss the following transformation properties considering a rectangular signal of width and height of 1 unit as shown in figure. (i) Time Shifting (ii) Time Scaling (iii) Time Reversal  	CO1
(b)	Find whether the following signals are periodic or not. Also, find fundamental time period if signal is periodic (i)$\boldsymbol{x(t) = 2cos(10t+1) - sin\,(4t-1)}$ (ii) $\boldsymbol{x(t) = 2cos60\pi t + sin\,50\pi t}$	
(c)	Determine whether the following signals are energy or power signals: (i) $x(t) = Asin(t)$ (ii) (ii) $x(t) = e^{a\|t\|}$ for a>0	
Q2	(20 marks)	CO2
(a)	Discuss Associative and Distributive property of the convolution integral	
(b)	If $x(t) * h(t) = y(t)$ then show that $x(at) * h(at) = \frac{1}{\|a\|} y(at)$	
(c)	Find the convolution of the signals f(t)= 3 cos (2t) and $g(t) = e^{-\|t\|}$	
Q3		CO3
(a)	If Fourier transform of *x(t)* is *X(ω)* then find the Fourier transform of the following signals (i) *x(at)* (ii) $\frac{dx(t)}{dt}$	

(b)	If Fourier transform of $x_1(t)$ and $x_2(t)$ are $X_1(\omega)$ and $X_2(\omega)$ then find the Fourier transform of convolution of $x_1(t)$ and $x_2(t)$.	
(c)	Find the Fourier Transform of rectangular pulse as shown below.  	
Q4	(20 marks)	**CO4**
(a)	Find the Laplace transform of the following signals (i)$t^2\,u(t-4)$ (ii) $(Sin\omega t)$u(t-T/2), where T is the time period of the sinusoidal signal (iii) $x(t-t_o)\,u(t-t_0)$, if Laplace transform of *x(t)* is *X(s)*	
(b)	Find Inverse of Laplace transform of the following functions (i) $F(s) = \frac{4s+5}{(s-1)^2\ (s+2)}$ (ii) $F(s) = \frac{2s^2-6s+5}{s^3-6s^2+11s-6}$	
(c)	If Laplace transform of *x(t)* is X*(s)* then find Laplace Transform of (i)$\int_0^t x(t)dt$ (ii) $\frac{x(t)}{t}$	
Q5		**CO5, CO6**
(a)	Discuss sampling theorem	
(b)	Determine the Nyquist rate of the following signals (i) $x(t) = Sin200\pi t$ (ii) $x(t) = Sin^2 200\pi t$ (iii) $x(t) = Cos(150\pi t) \times Sin(100\pi t)$	
(c)	Find the Laplace transform of (i) t sin (ωt) (ii) (ii) t^2 sin (ωt)	

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TEE–404

B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

DIGITAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write the statements of two De-Morgan's theorems. Prove both of them. (CO1)

(b) If $ax = bx$ and $ax' = bx'$, then prove that $a = b$

Find the binary equivalent of '-18' if number is represented in : (CO1)

(i) Sign magnitude from

(ii) 1's complement representation

(iii) 2's complement representation

2. (a) Draw the five variable karnaugh map. Assign all the squares to their respective Minterm numbers. (CO2)

P. T. O.

(b) Simplify the Boolean function F using the don't-care conditions d in "Sum of products" form : (CO2)

$$f(w, x, y, z) = \Sigma\,(0, 1, 6, 7, 14, 15) + \Sigma d\,(3, 4, 11, 12)$$

(c) Realize the ExOR gate function using (i) All NAND gates and (ii) All NOR gates. (CO3)

3. (a) Design a combinational logic circuit that converts" BCD to Excess - "code. Write the truth table. (CO3)

(b) Differentiate between a "Half Adder" and a "Full Adder". Design these circuits and draw their combinational logic diagrams. (CO3)

(c) What is a Magnitude comparator ? (CO3)

4. (a) What do you mean by an "Edge- triggered D- Flip Flop ? Draw it's circuit diagram that includes RC differentiator clock input, present and clear functions. (CO4)

(b) Describe working of a JK Master/slave flip- flop. Why are M/S flip-flop greatly popular in industries ? (CO4)

(c) Convert an SR flip-flop to a D-flip-flop, write characteristic table of D-flip-flop and the Excitation table of SR Flip-flop. (CO4)

5. (a) Draw the circuit diagram of a Ring counter. write it's working with the help of drawing it's output wave shapes. Enumerate it's applications. (CO5)

(b) Draw circuit diagram of a shift counter (Johnson Counter) Draw it's truth table and explain in brief it's working. What do you understand by illegal states. (CO5)

(c) Design a "2×3 mod-6" counter. Draw wave shapes and write the truth table describing it's working. (CO5)